THE FORMAL-BRANCH MAP PRESERVES MULTIPLICITIES WHEN THE BASE IS A G-RING

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original lemma 0DRB: sources/stacks-chow/tags.txt, source lines 17216–17216. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DR0: sources/stacks-chow/tags.txt, source lines 17205–17205. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07PV: sources/stacks-chow/tags.txt, source lines 9810–9810. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack locally of finite type over a locally Noetherian scheme S. Let $x_0 : \mathrm{Spec}(k) \to \mathcal{X}$ is a morphism where k is a field of finite type over S with image $s \in S$. If $\mathcal{O}_{S,s}$ is a G-ring, then the map of Lemma 0DRB preserves multiplicities.*

Proof. By Lemma 0DR0 we may assume there is a smooth morphism $U \to \mathcal{X}$ where U is a scheme and a k-valued point u_0 of U such that $\mathcal{O}^\wedge_{U,u_0}$ is a versal ring to $\mathcal{X}$ at x_0. By construction of our map in the proof of Lemma 0DRB (which simplifies greatly because $A = \mathcal{O}^\wedge_{U,u_0}$) we find that it suffices to show: the multiplicity of an irreducible component of U passing through u_0 is the same as the multiplicity of any irreducible component of $\mathrm{Spec}(\mathcal{O}^\wedge_{U,u_0})$ mapping into it.

Translated into commutative algebra we find the following: Let $C = \mathcal{O}_{U,u_0}$. This is essentially of finite type over $\mathcal{O}_{S,s}$ and hence is a G-ring (More on Algebra,

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Proposition 07PV). Then $A = C^\wedge$. Therefore $C \to A$ is a regular ring map. Let $\mathfrak{q} \subset C$ be a minimal prime and let $\mathfrak{p} \subset A$ be a minimal prime lying over $\mathfrak{q}$. Then

$$R = C_\mathfrak{p} \longrightarrow A_\mathfrak{p} = R'$$

is a regular ring map of Artinian local rings. For such a ring map it is always the case that

$$\text{length}_R R = \text{length}_{R'} R'$$

This is what we have to show because the left hand side is the multiplicity of our component on U and the right hand side is the multiplicity of our component on $\text{Spec}(A)$. To see the equality, first we use that

$$\text{length}_R(R)\text{length}_{R'}(R'/\mathfrak{m}_R R') = \text{length}_{R'}(R')$$

by Algebra, Lemma 02M1. Thus it suffices to show $\mathfrak{m}_R R' = \mathfrak{m}_{R'}$, which is a consequence of being a regular homomorphism of zero dimensional local rings. $\square$